

My Journey into Data Visualisation: Learning to Tell a Story with Data

This year, I had the opportunity to take part in the Pacific Data Visualization Challenge. Before starting, I knew very little about data visualisation, coding, or how to turn a large dataset into something that could tell a meaningful story.

Over several days, I went from looking at previous winning projects and wondering how they were made, to analysing real data, creating my own charts, and even building my first website. This blog is about what I learned along the way.

Learning from the 2025 Winners

Before starting our own project, I looked through some of the winning entries from 2025. This helped me understand what makes a data visualisation interesting and effective.

Echoes of the Unprotected Sea

The first thing that impressed me was its strong artistic style. The colours worked extremely well together, and the overall composition was beautiful.

However, I also felt that the design sometimes overshadowed the data. I found myself paying more attention to how the project looked than to what the data was actually telling me. This taught me an important lesson: a visualisation should be beautiful, but the design should always help the reader understand the data.

A Decade of Change: Renewable Energy Growth and Educational Shifts in the Pacific

I liked how simple and interactive the charts were. They were easy to explore and did not overwhelm the reader with information.

However, I found it difficult to understand the connection between renewable energy and education. Although similar colours were used to connect the two themes visually, I felt that the story connecting them could have been stronger.

Beyond the Lifeline

This was one of the projects I found particularly interesting. It was highly interactive and entertaining, while still communicating its information clearly.

My main suggestion would be to provide a short introduction before showing the main visualisation. As a new reader, I wanted a little more context about what I was about to explore and why it mattered.

Birth in the Pacific: Unequal Access to Skilled Care

I liked how concise and clear this project was. The theme was easy to identify and the message was straightforward.

At the same time, I thought the topic could have been explored in greater depth. Compared with some of the other projects, the story felt quite simple and there may have been opportunities to add another dimension to make it more engaging.

Looking at these projects helped me realise that a great data visualisation needs two things at the same time: a strong visual design and a strong story. Having only one is not enough.

Finding Our Story: Climate Change and Mental Health

After exploring the previous winners, our team began thinking about our own project.

We became interested in the relationship between climate change, natural disasters and mental health in Pacific communities. We explored several datasets, including the number of people directly affected by disasters, changes in surface temperature, and suicide mortality rates.

Eventually, I focused on a more specific question:

How do major disaster events relate to changes in suicide mortality rates in Pacific Island communities?

I thought this was interesting because natural disasters affect much more than buildings and infrastructure. They can disrupt families, communities, jobs and people's everyday lives. I wanted to see whether major disaster events were followed by noticeable changes in suicide mortality.

Working with the Data

This was the part where I began to understand how much work happens before a chart is created.

First, I had to clean the country names so that different datasets could be matched. I extracted the years, converted data from wide format into long format, and then identified the year in which the largest number of people were affected by disasters in each country.

For example, Fiji's largest event in our dataset occurred in 2016, when 633,584 people were recorded as being directly affected. Vanuatu's largest event occurred in 2020, affecting 246,802 people, while Tonga's largest event occurred in 2018, affecting 84,311 people.

I then combined this information with suicide mortality data. For each country, I compared the suicide mortality rate in the year before the major disaster, the year of the disaster, and the following year.

The results were not what I initially expected.

Fiji experienced an increase of 0.23 in the suicide mortality rate from the year before the disaster to the year after it. Kiribati increased by 0.30. However, Papua New Guinea, Solomon Islands, Tonga and Vanuatu all showed decreases over the same comparison period.

This led to one of the most important lessons from the project: data does not always tell the story you expect it to tell.

It would be tempting to say that natural disasters increase suicide mortality, but our data does not support such a simple conclusion. A more accurate interpretation is that:

Major disasters are associated with different changes in suicide mortality across Pacific Island countries.

There are probably many other factors involved, and a simple before-and-after comparison cannot establish that disasters caused the changes we observed.

Day 1: Where Do I Even Start?

Today I officially started working on the Pacific Data Visualization Challenge. It was my first time participating in a research project like this, and at first I was completely lost.

The data was unfamiliar. The visualisation framework looked professional and complicated. Even the research topic seemed difficult. I honestly had no idea where to begin.

So instead of immediately trying to create something, I started by looking at previous winning entries.

As I explored projects such as Echoes of the Unprotected Sea and Beyond the Lifeline, I gradually developed my own opinion about what makes a good project. I realised that our work should not only look attractive enough to catch people's attention, but also have a clear argument that makes the information easy to understand.

I did not produce anything today, but I still learned something important: sometimes the first step in creating your own work is learning how to look carefully at the work of others.

Day 2: Starting to Understand the Story

After the first day of exploring, and with lots of patient explanations from everyone around me, I finally began to understand the direction of our project.

The goal was not simply to make some charts. We needed to use data to tell a story.

Our broader idea was to explore natural disasters, environmental change and mental health across Pacific countries. Once I understood the logic behind the project, everything started to make much more sense.

I also began learning how to create charts with the help of AI tools. I learned how to import data, generate basic visualisations, change the structure of charts, work with templates and gradually understand how code controls what appears on the screen.

My first charts were definitely not beautiful. The layouts were awkward, the legends needed work, and the fonts were not quite right.

But when my first chart finally appeared correctly on the screen, I felt an enormous sense of achievement.

Day 3: Four Charts!

Today we made a major breakthrough.

With help from my brother and the rest of the team, I managed to complete four visualisations using AI. They covered natural disasters, rural population percentages, surface temperature and suicide mortality rates across the Pacific.

For the first time, I could actually see the different pieces of our project coming together.

There was still a lot of debugging. I had to fix small details, improve the resolution and make the charts easier to read. They were certainly not perfect, but they were usable.

More importantly, I was becoming much more comfortable with the process.

Only a few days earlier, I had looked at other people's visualisations and wondered how anyone could make something like that. Now I was starting to make my own.

Day 4: The Details Matter

When I got home today, I opened a chart that Veronica had helped me improve.

I was shocked.

It was the same data and basically the same chart, but it looked completely different. It was cleaner, more organised and much more professional than my original version.

That comparison taught me something that I don't think I would have understood just from reading a tutorial.

Data visualisation is not simply about putting data into a chart. Layout, spacing, font size, labels and overall cleanliness all matter. Small details can completely change how someone experiences the same information.

I still have a lot to learn from Veronica, but now I pay much more attention to those small details when making my own charts.

Day 5: Learning How to Talk to AI

Today I went to Veronica with an important question:

Why does her AI seem so much smarter than mine?

She explained that she gives the AI extremely detailed instructions. Instead of simply asking it to “make the chart look better”, she carefully describes exactly what needs to change and how she wants the final result to look.

I tried doing the same thing and quickly discovered that this was much harder than it sounded!

AI can misunderstand you in surprisingly creative ways. If your instructions contain a loophole, it somehow seems to find it. Sometimes I would fix one problem only for another one to suddenly appear.

It made me realise that working effectively with AI is actually a test of human communication. You have to understand what you want before you can explain it clearly to a machine.

The difference is not always the AI tool itself. The quality of the result depends heavily on the quality of the instructions, the thinking behind them and the experience of the person using the tool.

In other words, AI does not remove the need to think. It makes clear thinking even more important.

Day 6: I Built a Website!

Today I explored the website Veronica had created for the project. Seeing what she had built made me want to create one myself.

At first, I tried working from her template. After experimenting with it for a while, though, I decided to start again and see whether I could build something myself.

Eventually, I created my first simple website.

It is definitely not going to win any web design awards. The page is quite plain, there is too much text, and there are no exciting interactive features.

But it works.

And I made it.

That small achievement reminded me of how I felt on Day 1, when almost everything about this project seemed unfamiliar and difficult.

What I Learned

When I started this challenge, I thought data visualisation was mainly about making good-looking charts.

Now I understand that it is much more than that.

It involves asking an interesting question, finding appropriate data, cleaning and combining datasets, thinking carefully about what the results actually mean, designing visualisations that people can understand, and then connecting everything together into a story.

I also learned something about AI. AI can help me code, create charts and build websites much faster than I could on my own. But it cannot decide what question is worth asking or automatically know what I am trying to communicate. I still need to think carefully and explain what I want.

Most importantly, I learned not to be afraid of things I don't understand yet.

Six days ago, I did not know where to begin.

Now I have analysed real Pacific data, created visualisations, investigated a research question and built my first website.

There is still a lot I don't know, but that is probably the most exciting part.